

FEASIBILITY STUDY

A feasibility study is conducted to determine whether the proposed PilgrimIQ: Personalized Pilgrimage Decision Support System is practical, achievable, and beneficial before development begins. It evaluates the project from economic, technical, and behavioral perspectives to ensure that the system can be successfully implemented and accepted by its intended users.

1.Economic Feasibility

Economic feasibility evaluates whether the benefits of the project justify the development and operational costs. PilgrimIQ is designed using cost-effective and open-source technologies, which significantly reduce the overall implementation cost.

The project utilizes technologies such as React.js for the frontend, Node.js with Express.js for the backend, MongoDB as the database, and Python for AI-based medical report analysis, health risk prediction, and recommendation generation. OCR (Optical Character Recognition), Natural Language Processing (NLP), and Machine Learning (ML) libraries used for analyzing medical reports are open source and freely available. During the development phase, the project can be hosted locally, while cloud deployment can later be carried out using affordable platforms such as Vercel, Render, or AWS. Additional operational costs may include domain registration, cloud hosting, weather API integration, Google Maps API usage, and system maintenance.

The platform provides long-term value by improving pilgrimage safety, supporting informed travel decisions, reducing health-related risks, and assisting pilgrims with accommodation, transportation, food recommendations, and emergency services. Therefore, the overall benefits outweigh the implementation cost, making the project economically feasible.

2.Technical Feasibility

Technical feasibility determines whether the required technology, tools, and technical expertise are available to develop the system successfully.

PilgrimIQ can be implemented using modern web technologies such as React.js, Node.js, Express.js, MongoDB, Python, HTML5, CSS3, and JavaScript. AI functionalities such as medical report analysis, health risk prediction, travel readiness assessment, Pilgrim Safety Index (PSI) generation, crowd prediction, and personalized recommendation systems can be developed using Python libraries, OCR tools, NLP techniques, and Machine Learning algorithms. Authentication can be implemented using JWT, while external services such as Weather APIs and Google Maps APIs can be integrated to provide weather forecasts, navigation, nearby hospitals, accommodation, food facilities, and transportation information.

The required software tools include Visual Studio Code, Git, GitHub, Node.js, MongoDB, Python, Postman, and modern web browsers. Since these technologies are widely available, well-supported, and suitable for AI-enabled web applications, the project is technically feasible.

3.Behavioral Feasibility

Behavioral feasibility examines whether users will accept and effectively use the proposed system.

PilgrimIQ is designed with a simple, user-friendly, and responsive interface that can be easily used by individual pilgrims, family groups, general physicians, and system administrators. The platform provides AI-assisted health assessment, travel readiness evaluation, personalized recommendations, accommodation suggestions, transportation guidance, food recommendations, and intelligent journey assistance, making pilgrimage planning safer, easier, and more convenient.

The system prioritizes user privacy and data security by implementing secure authentication, encrypted storage of sensitive health information, and role-based access control. Since the platform addresses real-world challenges such as health risk assessment, travel planning, emergency support, and personalized decision-making during pilgrimages, users are expected to adopt the system with minimal training. Hence, the project is behaviorally feasible.